

Chapter 53 — Surgery of the Ureter in Children: Ureteropelvic Junction, Megaureter and Vesicoureteral Reflux — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Gold standard UPJO (ureteropelvic junction obstruction) repair:** Anderson-Hynes dismembered pyeloplasty, about 95% success.
- **Paediatric hydronephrosis:** AP (anteroposterior) pelvis 12 mm; dilatation ≠ obstruction (Koff).
- **Moiety ≤5–10% function** → partial nephrectomy, not repair.
- **Tunnel length:diameter 5:1** (Paquin), for reimplant and megaureter alike.
- **Commonest intravesical reimplant:** Cohen cross-trigonal (>98%).
- **Lich-Gregoir:** voiding dysfunction up to 10%, especially bilateral.
- **Megaureter** ≥7 mm; normal ≤5.0–6.5 mm.
- **POM (primary obstructive megaureter) surgery:** DRF (differential renal function) <40% or fall ≥5%, recurrent UTI (urinary tract infection), progressive dilatation.
- **POM <10 mm** virtually never operated; ≥10 mm → complications, surgery.
- **Most used bulking agent:** Deflux (Dx/HA, dextranomer/hyaluronic acid), FDA (US Food and Drug Administration) approval 2001.

URETEROPELVIC JUNCTION OBSTRUCTION

Cut-offs and epidemiology:

- **Antenatal scan about 20 weeks;** urogenital = commonest anomaly; dilatation commonest within it.
- **Male, left; bilateral 10–40%.**
- **Associated urological anomaly up to 50%;** commonest = contralateral UPJO (10–40%).
- Then **dysplasia** → MCDK (multicystic dysplastic kidney) → agenesis; also horseshoe kidney, VATER.
- **VUR up to 40%** with UPJO; high spontaneous resolution.
- **About one-third need surgery;** most resolve.

Cause by age:

- **Infant** → **intrinsic** (circular muscle defect); **older child** → **crossing lower-pole vessel**, intermittent pain and vomiting.

Operate if:

- **Rising AP diameter · low or falling DRF · breakthrough UTI · pain.**
- **Infant: failure to thrive, feeding difficulty, constipation.**

Gold standard and technique:

- **Infant** → **open** (incision ≈ port size); open → **extraperitoneal**.
- **Anderson-Hynes:** originally for **retrocaval ureter**; preserves vessel, excises adynamic segment.
- **Spatulate laterally;** stenotic segment = handle, excised last.
- **6-0/5-0 absorbable;** tense pelvis → **21 G needle** decompression first.
- **Bipolar** for bleeding; bleeding = **commonest conversion reason**.
- Pyeloplasty success **90–100%**; “success” definitions vary.
- Stent duration **1–4 weeks** → no outcome difference.

Alternatives:

- **Foley Y-V:** high insertion, **no crossing vessel**.
- **Spiral flap:** long stricture; **length:width ≤3:1**.
- **Vertical flap:** box-shaped extrarenal pelvis.
- **Heineke-Mikulicz** inferior to dismembered.
- **Vascular hitch** (Hellström 1951): **anterior lower-pole vessel + NO intrinsic obstruction**; normal peristalsing UPJ; no calyceal dilatation.
- **Uretero-calicostomy:** scarred short ureter + dilated lower calyx, thin parenchyma.

Access:

- **Posterior lumbotomy:** infants and toddlers only; prone.
- **Flank:** supine, tip of 11th–12th rib.
- **Transperitoneal 45–60° flank;** left → **transmesocolic**.

- **Retroperitoneal 80–90°;** table **not flexed**; 200–300 mL space.
- **No retroperitoneal route** for horseshoe or retrocaval ureter; ideal for duplex lower-moiety robotic repair.
- **Robot:** shorter learning curve than laparoscopy.
- **Redo pyeloplasty** → **transperitoneal**.

Endopyelotomy:

- **Balloon under 18 years: poor;** secondary (scar) UPJO > primary.
- **Niche = failed pyeloplasty** → double-J 6 weeks; **redo pyeloplasty still best**.

Stents:

- **Blue stent:** open 12–48 h, knotted, **out at 7–14 days, no anaesthetic**.
- **Foley 24–48 h only with double-J** (stent-induced reflux).
- **Emergency diversion before pyeloplasty** → **nephrostomy**, not retrograde stent.
- **String on double-J** → more febrile UTI.

Other ureteric lesions:

- **Congenital stricture:** mid-ureter, 1–3 cm → ureteroureterostomy.
- **Polyp:** upper third, male, left; **fibroepithelial**; confirm by RPG (retrograde pyelography) or ureteroscopy.
- **Retroperitoneal fibrosis:** steroids → azathioprine → double-J → ureterolysis.

VESICoureTERIC REFLUX (VUR)

Principles:

- **Treat secondary reflux first;** length:diameter 5:1; straight course.
- **Pfannenstiel 2 cm above symphysis;** **vertical incision for redo**.
- Bladder opened to **2 cm of neck**; orifice **5-Fr feeding tube**; circumscribe 1–2 mm.

Eponyms:

- **Politano-Leadbetter (1958):** suprahiatal, new superomedial hiatus; long tunnel; high grade.
- **Glenn-Anderson (1967):** infrahiatal → bladder neck; least kinking, short tunnel.
- **Cohen (1977):** cross-trigonal; commonest; trabeculated bladder; **hard later retrograde access**.
- **Lich (1961)–Gregoir (1964):** extravesical; **no haematuria or spasm**; 5-cm tunnel.
- **Zaontz modification** (around UVJ) → nerve injury; **abandoned**.

Numbers:

- **Pelvic plexus 1.5–2 cm dorsomedial to UVJ** (ureterovesical junction) → hug ureter, stop at UVJ.

- **Open reimplant success 95.9%:** AUA meta-analysis (Elder 1997).
- Low-grade primary VUR → reimplant **about 100%**.
- **Ultrasound 6–12 weeks** postoperatively (AUA 2010, amended 2017).
- **RALUR-EV** (robotic extravesical reimplant): low lithotomy, 12-mm Hasson umbilical port, two 8-mm ports, Y-detrusorotomy 4–5 cm.

Postoperative:

- **Omit VCUG** (voiding cystourethrogram): low-grade, normal ultrasound, no dysfunction or UTI.
- **Do VCUG:** high grade, dysfunctional bladder, new hydronephrosis or UTI.
- **Early persistent reflux** → usually resolves by 1 year.
- **Contralateral VUR** about 1 in 10, mostly resolves; risk **age <6 y, capacity <50%, duplex**.
- Asymptomatic contralateral VUR **under 4–5 y → prophylaxis**.
- Postoperative dilatation ≤ **maximal reflux dilatation** = acceptable.
- **Acute obstruction within 2 weeks** → double-J or nephrostomy; internalise early (“**dry reimplant**”).
- **Late angulation:** neo-hiatus too **lateral or anterior**; kinks on filling.

Redo:

- **Psoas hitch one side only**, before the tunnel, non-absorbable sutures.
- Both short → **hitch + TUU (transureteroureterostomy)**.
- **Boari flap:** based on the posterior wall.
- **Monti-type bowel ureter:** no tapering, no metabolic cost.

ENDOSCOPIC TREATMENT OF VUR

- **Matouschek 1981; STING** (subureteric Teflon injection), O'Donnell and Puri **1986**.
- **Teflon never FDA-approved:** migration (90% <40 µm).
- Migration: <50 µm intravascular; <80 µm phagocytosed.
- **Deflux 80–250 µm;** loses about 23% volume; low-grade single injection about 90%.
- **Vantris 300 µm**, non-degradable; higher resolution; **late obstruction**; not FDA-approved.
- **Swedish Reflux Trial:** 20% recurrence at 2 years after Dx/HA.
- **Needle 3.7–5 Fr**, bevel up, **6 o'clock**; mound by **0.1–0.2 mL**.
- **STING:** 2–3 mm below orifice; **HIT** (hydrodistension implantation technique) intramural > STING; **double HIT** best.
- **Volcano mound** → best success.
- Bleeding → pressure, **never cautery**.
- Follow-up: **antibiotics 3 months**; ultrasound **2–6 weeks**; 1–2 reinjections, then surgery.

- Recurrence risk: **febrile UTIs, BBD (bladder and bowel dysfunction), scarring**.

MEGAURETER

- **“Megalooureter”:** Caulk 1923.
- **4 types:** obstructed · refluxing · non-obstructed non-refluxing · refluxing + obstructed.
- **POM:** adynamic distal segment; craniocaudal maturation; interstitial cells of Cajal.
- **POM 10–23%** of antenatal dilatations; male, left; **bilateral 25%**; contralateral anomaly 10–15%.
- **Renogram T½ unreliable:** false positive up to 44%.
- Follow into **early adulthood** (worsening at puberty).
- **Infant temporising:** distal cutaneous ureterostomy, refluxing reimplant, nephrostomy, dilatation and stent.
- **Plication or folding** (Starr, Kalicinski) only **<1.75 cm**; 8–10-Fr catheter.
- **Hendren excisional tapering (1969):** massive ureter; ischaemia risk; running proximal two-thirds, interrupted distal third.
- **Taper gradually;** stent: feeding tube 7 days or double-J 4 weeks.
- **POM outcome >90%;** better than refluxing megaureter.
- **Dilatation and stenting:** 70–80%; double-J 2–6 months; complications up to one-third.

CLASSIC TRAPS

- **Trap:** dilatation = obstruction — **Actually:** obstruction = progressive damage (Koff).
- **Trap:** laparoscopy for infant UPJO — **Actually:** open preferred in infants.
- **Trap:** vascular hitch for any crossing vessel — **Actually:** only without intrinsic stenosis.
- **Trap:** endopyelotomy first-line in children — **Actually:** reserve for failed pyeloplasty.
- **Trap:** retroperitoneal route for horseshoe kidney — **Actually:** contraindicated (posterior approach).
- **Trap:** bilateral psoas hitch — **Actually:** one side only, plus TUU.
- **Trap:** reimplant secondary (neurogenic or valve) reflux — **Actually:** treat bladder first (anticholinergic + CISC).
- **Trap:** routine bilateral reimplant to prevent contralateral VUR — **Actually:** most resolve spontaneously.
- **Trap:** prolonged renogram T½ proves megaureter obstruction — **Actually:** up to 44% false positive.
- **Trap:** plicate any megaureter — **Actually:** folding only <1.75 cm; excise larger.